

LLM Routing Layer: Natural Language → Structured JSON

User Input:

"A 3-month European call option on SPY,
strike \$740, sigma around 0.18, use Heston.
vol-of-vol is 0.3, correlation -0.7."

DeepSeek LLM

LLM Output (JSON):

```
{  
  "model": "heston",  
  "option_type": "call",  
  "S": 740.0, "K": 740.0,  
  "T": 0.25, "r": 0.043,  
  "sigma": 0.18, "beta": 1.0,  
  "kappa": 2.0, "theta": 0.04,  
  "xi": 0.3, "rho": -0.7,  
  "v0": 0.04  
}
```

Unified PINN → Option Price: \$34.52
Greeks: $\Delta=0.582$ $\Gamma=0.0031$ $\nu=127.4$
(inference time: ~2 ms on GPU)

*Model selection rules: σ only → BSM | β mentioned → CEV
 $\kappa, \theta, \xi, \rho$ present → Heston | default → BSM with $\sigma=0.2$*